DREAM: Delay-Resistant Acceleration with Momentum

Optimal Acceleration under Delayed Gradients

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Delayed Gradient Setup

$$\min_{x \in \mathbb{R}^d} f(x) \qquad f \text{ is } L\text{-smooth and } \mu\text{-strongly convex}, \qquad \kappa = \frac{L}{\mu}.$$

Without delay

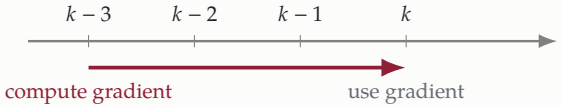
Gradient descent

$O(\kappa)$

Accelerated methods

$O(\sqrt{\kappa})$

Delayed setup



At round k, use the gradient queried τ iterations earlier.

Can we retain acceleration when gradients are delayed?

Optimization Under Delay

Delayed Gradient Descent

$$x_{k+1} = x_k - \alpha \nabla f(x_{k-\tau})$$

Uses a stale gradient every iteration.

"Idle" Methods

Request → Wait → Update

Waits for the requested gradient.

Cost of Delay

For fixed delay τ, the delayed first-order lower bound implies

$$\Omega\big((\tau+1)\sqrt{\kappa}\log\tfrac{1}{\varepsilon}\big)$$

for small relative accuracy.

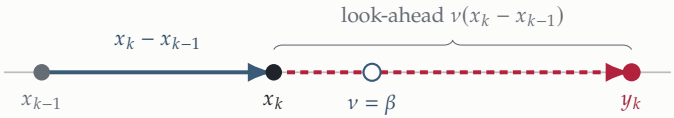
Can we achieve accelerated convergence under fix delays, exact gradients and no idle rounds?

Arjevani–Shamir–Srebro (2020): delayed first-order span model.

Generalized Momentum: Two Independent Parameters

Generalized Momentum Method (GMM)

$$x_{k+1} = x_k - \alpha \nabla f(y_{k-\tau}) + \beta(x_k - x_{k-1}), \qquad y_k = x_k + \nu(x_k - x_{k-1}).$$



Heavy-ball: $\nu = 0$ **Nesterov:** $\nu = \beta$ **GMM:** ν is free

Key idea: Look-ahead ν can be tuned independently of momentum β.

GMM: Gürbüzbalaban et al. (2023); Van Scoy et al. (2018) for $\tau = 0$.

Limits of Classical Momentum on Quadratics

Setting: Strongly convex quadratics, $f(x) = \frac{1}{2}x^\top Q x - b^\top x, \quad \mu I \preceq Q \preceq LI.$

Theorem (Gürbüzbalaban, Rafi, Aybat, Bajwa 2026)

For every fixed positive delay, the best achievable spectral gaps of stable HB and NAG satisfy the following asymptotic upper bounds:

Method	Delay	Upper bound on $1-\rho$
Heavy-ball	$\tau \geq 1$	$O_\tau(\kappa^{-1})$
Nesterov	$\tau = 1$	$O_\tau(\kappa^{-2/3})$
Nesterov	$\tau \geq 2$	$O_\tau(\kappa^{-1})$

$\underbrace{\kappa^{-1/2}}$ accelerated gap vs. $\underbrace{\kappa^{-2/3},\ \kappa^{-1}}$ delayed HB/NAG bounds

No stable retuning of HB or Nesterov recovers $\sqrt{\kappa}$-acceleration under delay.

DREAM: Proposed Method

Delay-matched parameters ($M = 2\tau + 1$)

$$\alpha = \frac{1}{M^2 L}, \qquad \beta = 1 - \frac{2}{M\sqrt{\kappa}}, \qquad \nu = 2\tau + \frac{1}{2}$$

Theorem (Gürbüzbalaban, Rafi, Aybat, Bajwa 2026)

Under the DREAM parameter choices above, for fixed delay τ and sufficiently large κ, the iterates converge linearly:

$$\|x_k - x_\star\| \le C \left(1 - \frac{c}{M\sqrt{\kappa}}\right)^k, \qquad M = 2\tau + 1.$$

No idle rounds: $\widetilde{O}_c\left((\tau+1)\sqrt{\kappa}\log\frac{1}{\varepsilon}\right)$ stale-gradient updates.

Why Look-Ahead Matters

1. Characteristic roots determine the speed

Let z_i denote the characteristic roots of the delayed GMM recursion.

$$\rho = \max_i |z_i| \quad \Longrightarrow \quad \text{asymptotic convergence factor } \rho^k$$

Smaller ρ means faster asymptotic convergence.

2. The slow-root expansion reveals the effect of delay

$$|z_\pm|^2 = 1 + (\tau - \nu)\delta + O(\delta^{3/2})$$

First $\beta \to 1$; then $\delta \downarrow 0$, for roots near $z = 1$.

3. Move the slow roots inside the unit circle

$$|z_\pm| < 1 \quad \Longleftarrow \quad \nu > \tau$$

DREAM uses $\nu = 2\tau + \frac{1}{2} > \tau$: look-ahead counteracts delay.

Beyond Quadratics: A Control-Theoretic Approach

Quadratic objectives

$$\nabla f(y) = Qy - b$$

Linear gradient

Characteristic-root analysis

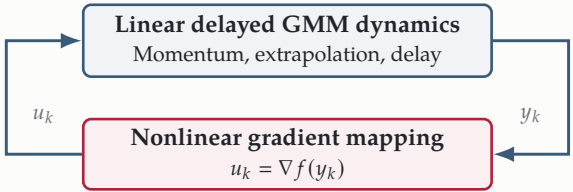
General smooth, strongly convex functions

$$u = \nabla f(y)$$

Generally nonlinear gradient

Characteristic roots alone are insufficient

Our proof approach (Gürbüzbalaban et al., 2026)

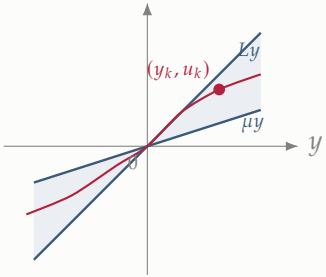


Decompose DREAM into linear dynamics and nonlinear gradient feedback.

Understanding IQCs

At each iteration

$u = f'(y)$



$$(u_k - \mu y_k)(L y_k - u_k) \geq 0$$

Pointwise gradient restriction (1D)

Across iterations

(y_{k-2}, u_{k-2}) → (y_{k-1}, u_{k-1}) → (y_k, u_k)

Weighted quadratic terms
within and across iterations

Integral Quadratic Constraint (IQC)

$$\sum_{k=0}^{N} \sigma_k \geq 0$$

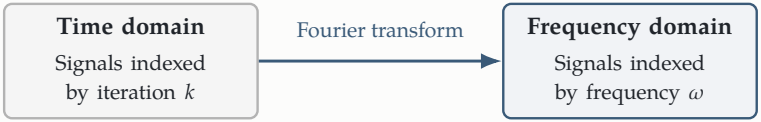
for every finite horizon N.

IQCs capture gradient relationships across iterations, beyond pointwise bounds.

Zames–Falb (1968); Lessard–Recht–Packard (2016).

From Time Domain to Frequency Domain

IQCs constrain signals over time. Can we study these constraints more easily?



Parseval's Identity

$$\sum_{k=0}^{\infty} \|v_k\|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \|\widehat{v}(e^{i\omega})\|^2 \, d\omega$$

Here, v_k is any square-summable signal; the Fourier transform preserves its energy.

Frequency-domain analysis preserves signal energy and simplifies the study of linear dynamics.

Proof Strategy: Certifying Accelerated Convergence

Lemma 5.2 — Convergence Rate Lemma (Gürbüzbalaban, Rafi, Aybat, Bajwa 2026)

To certify convergence at rate ρ, we verify two conditions:

H1: Linear Stability

The rate-weighted delayed linear system is stable.

H2: Frequency-Domain Inequality

A strict frequency-domain inequality controls the nonlinear feedback.

$\Downarrow$

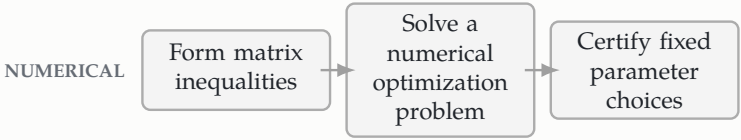
Energy Dissipation Certificate

Together with the gradient IQC, these conditions control a suitable energy measure over time.

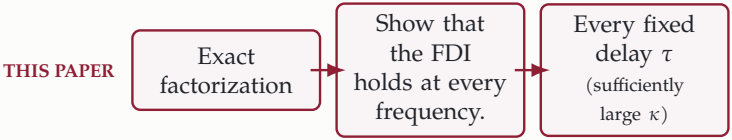
$\Downarrow$

Accelerated convergence: $\|x_k - x_\star\| \le C\rho^k$.

Analytic Verification of the FDI



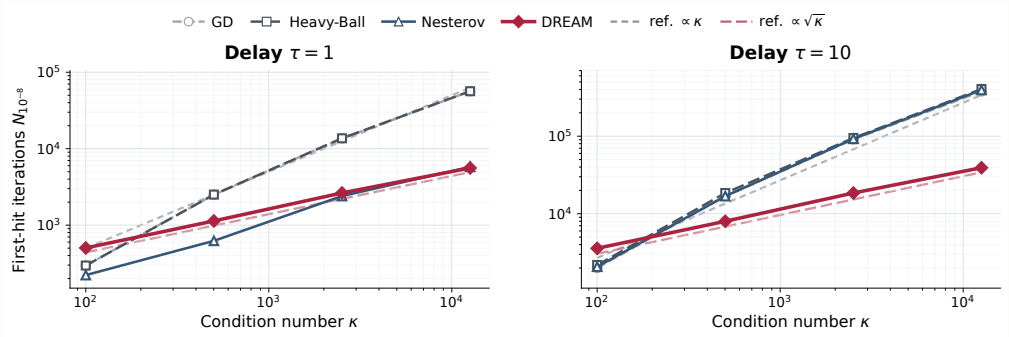
Can we certify the FDI without a numerical search?



Analytic certification replaces per-delay numerical searches.

Numerical Experiments (Strongly Convex)

Nonlinear test: $f(x) = \frac{1}{m}\sum_{i=1}^{m} \log(1 + e^{-y_i a_i^\top x}) + \frac{\mu_r}{2}\|x\|^2, \quad (m, d, \mu_r) = (1000, 10, 0.1).$



$N_{10^{-8}}$: first-hit relative-distance iterations. GD/HB/NAG rate-tuned; DREAM closed-form.

DREAM retains near-$\sqrt{\kappa}$ scaling on a non-quadratic objective under delay.

Takeaways

1. **Classical momentum fails under delay.**
2. **DREAM restores near-optimal acceleration.**
3. **Analytic convergence guarantees beyond quadratics.**

Paper

Gürbüzbalaban, Rafi, Aybat, Bajwa. *DREAM: Delay-Resistant Acceleration with Momentum*, 2026.

Thank you!

Backup: Beyond Strong Convexity: Catalyst + DREAM

Extension: $f \in \mathcal{S}_{0,L}$, $\arg\min f \neq \varnothing$; fix $\tau \geq 1$, $c \in (0, 2/3)$.

Catalyst + DREAM

Regularize each outer subproblem:

$$H_s(x) = f(x) + \frac{\lambda}{2}\|x - v_{s-1}\|^2$$

$$\chi_\lambda = 1 + \frac{L}{\lambda}.$$

DREAM solves each H_s using **stale-gradient updates**.

High-accuracy complexity

Let $\bar{\kappa}_0 = \max\{2, \kappa_0(\tau, c)\}$, $\lambda = L/(\bar{\kappa}_0 - 1)$, and $R_\star = \|x_0 - x^\star\|$.

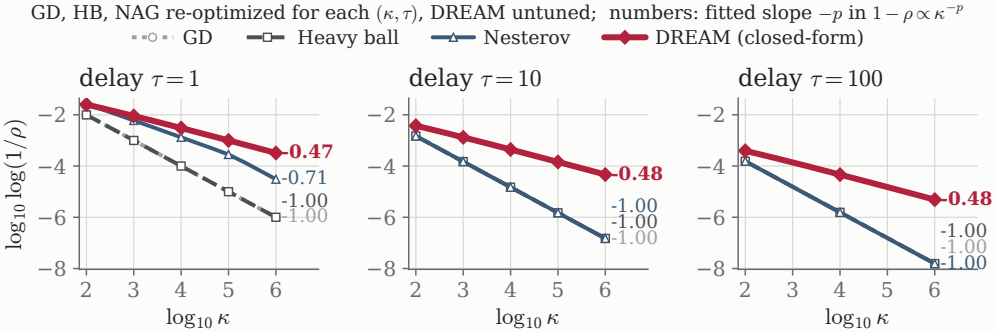
For $\varepsilon \leq \lambda R_\star^2$:

$$N_\varepsilon = \widetilde{O}_c\left((\tau + 1)\sqrt{\frac{LR_\star^2}{\varepsilon}}\right).$$

Catalyst outer guarantee: $f(x_s) - f^\star \leq \dfrac{4\lambda R_\star^2}{(s+1)^2}$.

Near-optimal delay–accuracy scaling without strong convexity.

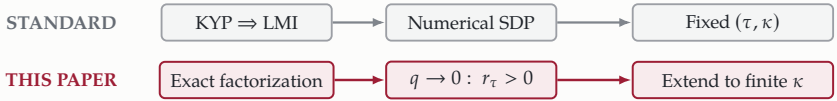
Backup: quadratic rates for $\tau = 1, 10, 100$



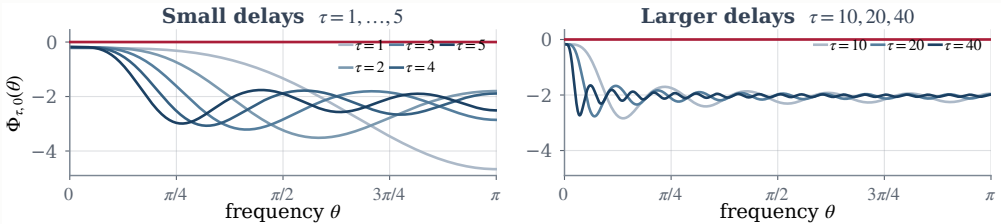
Worst-case spectral radius over $\lambda \in [\mu, L]$; GD, HB, NAG re-optimized per (κ, τ) (coarse-to-fine grid search on (α, β)); DREAM with the closed-form parameters of Result 2. Delay lowers DREAM's constant (the curves shift down with τ) but never its slope.

Backup: An Analytic Certificate for Every Delay

Goal (H2): $\Phi_{\tau,q}(\theta) < 0$ for every frequency θ, $\quad q = \kappa^{-1/2}$.

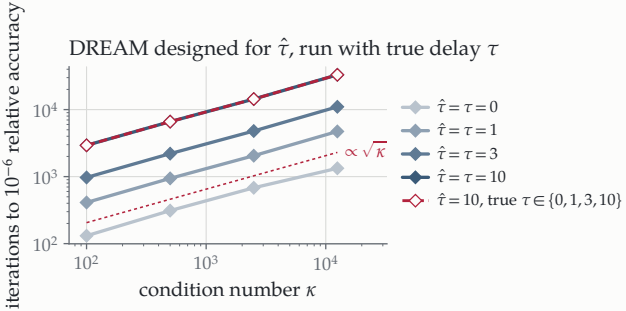


$$\Phi_{\tau,0}(\theta) = -\frac{r_\tau(\cos\theta)}{4M^2}, \qquad r_\tau(\cos\theta) = c_0 + 2\sum_{m=1}^{2\tau} c_m \cos(m\theta), \quad M = 2\tau + 1.$$



All-delay sign: $r_\tau(x) > 0$ on $[-1, 1]$ for all $\tau \geq 1$ (Dirichlet away from 0; Taylor–Faulhaber near 0).
Finite κ: low-frequency extension yields strict FDI — no per-delay SDP.

Backup: robustness to an overestimated delay



Parameters designed with $M = 2\hat{\tau} + 1$ for a conservative bound $\hat{\tau} = 10$, while the recursion runs with the true delay $\tau \in \{0, 1, 3, 10\}$ (logistic regression, $\varepsilon = 10^{-6}$).

Overestimating the delay keeps the $\sqrt{\kappa}$ scaling and costs a κ-independent factor set by $\hat{\tau}$ (the four $\hat{\tau} = 10$ runs are indistinguishable): one needs an upper bound on τ, not τ itself.